

A Machine Learning Approach for an Early Prediction of Preterm Delivery

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Abstract—The preterm birth presents a major cause of the infants' deaths, or the consequent health impairments globally, with an increasing trend of the preterm rate. The enormous global burden on both families and society calls for the preventive and predictive measures. The electrohysterogram (EHG), electrical activity of uterus as measured by surface electrodes, is a noninvasive and affordable tool for effective monitoring of both pregnancy and labour. In this study, the possibility of an early prediction of preterm delivery from the EHG recordings made between 22nd and 25th week of the gestation is explored. A set of novel features, including those exploiting signal's nonstationarity, based on the predictive modelling, and empirical mode decomposition, was evaluated on 15min long EHG recordings from the publicly available Term-Preterm EHG (TPEHG) database. On average, Random Forest classifier combined with artificial sampling, tested using 10-fold cross-validation on 322 samples (38 preterm) provided for 99.23% accuracy, with 98.40% sensitivity, and area under curve of 99%. The proposed approach has an additional advantage achieving the classification improvement over shorter, 15min long EHG recordings.

I. INTRODUCTION

A preterm birth, defined as a live born delivery before the 37th week of gestation, presents a global health and social burden, with more than 1 in 10 babies born preterm worldwide [1]. As the main cause of infant death and potential life-long health impairment, the preterm birth adversely affects both families and society [2,3]. The numerous risk factors: clinical, biological, social, and economic contribute in a complex interplay to preterm delivery [4]. A prediction of a preterm birth allows for an adequate, and timely preventive medical intervention. Predicting premature labour is not an easy task, for both multiple potential risks in interaction, and insufficient knowledge on the uterus physiology [5].

The electrohysterogram (EHG) is an abdominal surface recording of the uterine electrical activity. The electrical activity of uterus provides valuable information on its contraction [6,7]. The animal studies show that the relevant information obtained by the electrodes directly placed on a uterus transfer to the externally recorded surface EHG, both in pregnancy and labour [8,9]. The use of EHG to monitor and predict the risk of the preterm labour is subject of many studies. An attempt to standardize the EHG interpretation was made in [10], whereas in [6] a mathematical model was developed to help to address the problems associated with abdominal EHG recordings, and support interpretation of the obtained results.

The uterus activity increases gradually as the pregnancy develops, and the number of electrical connections

between the cells increases in preparation for the delivery [4]. The significant changes in EHG recordings can be noticed three to four days before delivery [11]. Thus, EHG can be successfully used as a non-invasive and cost-effective diagnostic tool for monitoring and detection of the labour onset. The EHG recorded bursting activity corresponds to the burst measurements obtained by the clinically used devices: tocodynamometer, and intrauterine pressure catheter [4].

This paper focuses on a more demanding task: prediction of the pregnancies resulting in a preterm delivery from EHG recordings made between the 22nd and 25th week of gestation. The publicly available Term-Preterm EHG Database (TPEHG) was used [5]. The studies confirming the potential of EHG as an early diagnostic tool exist [4,5,12]. As summarized in [5], many studies use only burst EHG segments characterized by their spectral features: the peak and median frequency, the mean and deviation of burst's power spectral density (PSD), and the number and duration of bursts. The analysis of the whole EHG recording in [13] included signals' root-mean square (RMS) and the median frequency (MF) of a signal's spectrum. The prognostic value of the nonlinear indices in EHG: sample entropy, correlation dimension, and Lyapunov exponent, was first evaluated in [5], and the statistically significant changes were observed in the estimated sample entropy.

The performance of several machine learning algorithms for an early prediction of the preterm birth was evaluated in [4], over the same dataset as in this paper. The polynomial classifier with four features: the peak and median frequency, RMS, and the sample entropy estimated over one 30 min long differential EHG channel, achieved 96% sensitivity for the preterm class, 90% specificity, and 95% area under the receiver operating characteristic (AUC) [4].

Relying on these encouraging results, in this research the possibility to achieve good classification performance for shorter, 15 min long recording sessions, was investigated. Additionally, all available differential channels were used, and a new feature set was designed in an attempt to improve the sensitivity for a preterm class, preserving the high specificity. Several classifiers were tested (k-Nearest Neighbours, Support Vector Machine, Random Forest), augmented with Adaptive Synthetic Sampling Approach [15] to provide for the class imbalance.

With respect to the nonstationarities in the 15 min long EHG recordings, empirical mode decomposition (EMD) [15] of EHG channels was evaluated. The EMD enables identification of the signal components - Intrinsic Mode Functions (IMF) in the frequency range of interest, and

using IMF's Hilbert transformation provides the time-frequency spectra for monitoring changes in the energy distribution over frequency in time. The features from the relevant IMFs and from the Hilbert energy spectra were estimated. The Hilbert spectra of IMFs has already been used for the EHG feature derivation in [16], with the maximum amplitude as a single feature for each IMF. The feature extraction was done only on the burst segments from EHG recording made in the third trimester of the pregnancy and 24h before the labour. In this study, multiple features were extracted from the IMFs belonging to the frequency range [0.3-1.5Hz], both in time and spectral domain, over the whole EHG recording made in the second trimester.

II. MATERIALS AND METHODS

A. EHG Database and Preprocessing

The Term-Preterm EHG Database is publicly available at Physionet [17]. The recordings were obtained during regular examinations around the 22nd and 32nd week of gestation at the University Medical Centre Ljubljana, Department of Obstetrics and Gynecology [5]. The details on electrode placement and recording protocol are provided in [5]. In brief, the four electrodes were placed on the abdomen, symmetrically, above (*E1* and *E2*), and below (*E3* and *E4*) the navel, 7 cm apart. Each 30 min long recording session contains three differential signals: the top channel 1 (*E2-E1*), the left channel 2 (*E2-E3*), and the bottom channel 3 (*E3-E4*). All three channels of the

EHG recordings are filtered in analogue domain with the Butterworth low pass filter with cut-off frequency of 5 Hz. The signals were discretized at 20 Hz and subsequently filtered with a digital Butterworth filter with the passband [0.3 – 3] Hz [5].

From the 300 available EHG recordings, the focus was on the recordings made in an early stage of pregnancies, before the 26th week of gestation [5]. From the pregnancies with term deliveries (Early Term - ET group), altogether 143 recordings were available. Only 19 recordings were available in the preterm group (Early Preterm - EP). An example of a term and a preterm EHG, recorded in an early stage, is presented in the top panels of Fig. 1 a) and b), respectively.

The average patient statistics is the following. The women in the ET group were of age 29.7 years, EHG recordings were made around 22nd week, and delivered after 39.7 weeks. In the EP group, the mean age was 29.6 years, recorded in the 23rd week, with delivery after 34.2 weeks.

At the beginning of the recordings, the artefacts due to the electrode placement or patient movements were noticed, and for these reasons first 700 samples were removed from the further analysis. The occasionally occurring high-amplitude artefacts were removed by thresholding. An adequate threshold was selected by analyzing the amplitude histograms of the recordings. The signals are subsequently segmented in two parts of the equal duration and sent to the feature extraction and classification pipeline.

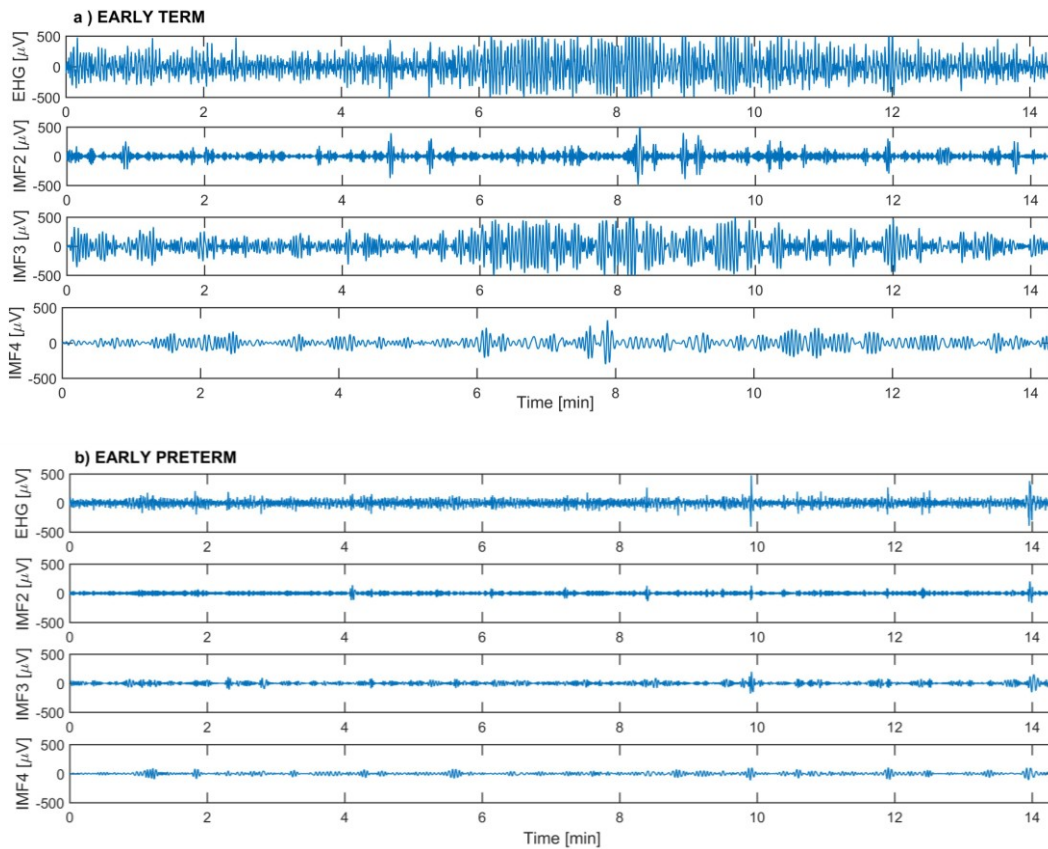


Figure 1. The EHG recordings from the 22nd week of gestation in a pregnancy with a) a term delivery (top panel), and b) a preterm delivery (top panel). The three bottom panels in a), and b) are the corresponding IMFs within a frequency range of interest: IMF2, IMF3 and IMF4, from top to bottom.

B. Feature Extraction

The first set of features was estimated for each channel of every recording. As an estimate of the signal energy, the root-mean square of each channel was calculated. The signal nonstationarity is exploited, and the changes of the features estimated from 30s long segments with 25% overlap were inspected. The EHG power spectral density was estimated using the Welch periodogram over the segmented recording. Additionally, the changes in the signal variance in 30s segments were observed.

The zero-crossing rate of each EHG channel and the first minima of the autocorrelation function were used to estimate the rate of the sign changes, and the correlation lag among the samples, respectively. Because of the EHG nonstationarity, these measures were calculated over 30s segments and represented in the feature matrix with their mean and variance. The Burg method was used to fit an autoregressive 10th order polynomial model to the EHG recordings. The model error was used as another feature estimated for all three channels of each recording separately.

Another set of features is comprised of the EMD related features. In each recording, every channel is decomposed into a number of components using the empirical mode decomposition [15]. The EMD is a data-driven decomposition resulting in a finite number of the frequency and amplitude modulated signals, intrinsic mode functions (IMF). The IMFs have zero-mean and limited frequency range. The first extracted IMF contains the fastest oscillations from the signal, the last extracted IMF the slowest, while the residuum, $r(t)$, represents the signal trend:

$$EHG(t) = \sum_{i=1}^N IMF_i(t) + r(t) \quad (1)$$

The obtained IMFs have a well-behaved Hilbert transformation providing their analytic representation using the instantaneous amplitude $A_{IMFi}(t)$ and phase $\phi_{IMFi}(t)$:

$$IMF_i(t) = \text{Real}\{A_{IMFi}(t)e^{j\phi_{IMFi}(t)}\} \quad (2)$$

Inserting the analytic form from (2) into (1), for each EHG channel, the Hilbert time-frequency energy spectra (HES), $H(f,t)$, can be obtained, representing the square of amplitude as a function of frequency and time [15].

The frequency content of the initial recording, due to the preprocessing filtering step, is limited to the [0.3-4] Hz range [5], removing the respiratory-related artefacts. Following the recommendations from [4], the reduced frequency range [0.3-1] Hz was observed, in order to avoid heart rate influence. As IMFs are frequency limited, but as well frequency modulated, the median frequency of IMFs was estimated. The results indicated that only the IMF_2 , IMF_3 , and IMF_4 belong to the recommended range. The selected IMFs for one term and one preterm channel are illustrated in Fig. 1, a) and b) respectively.

For each of the selected IMFs median frequency and the inter-quartile range of the differential $f_{IMFi}(t)$ are calculated. Using a median absolute deviation (MAD), the outlier thresholds were set to $3 \cdot 1.48 \cdot \text{MAD}$ both in $A_{IMFi}(t)$ and in differential $f_{IMFi}(t)$. These thresholds can be interpreted as the measures of scale. The power in [0.3-1], [1-2], and [2 – 3] Hz was estimated from the HES for each

EHG channel by integrating HES (f,t) in those frequency ranges and dividing by the recording time.

C. Classification algorithms

The classification of the recordings was performed using supervised machine learning algorithms: k nearest neighbours (kNN), Random Forest (RF), and Support Vector Machines (SVM). As a non-parametric procedure, kNN was selected for its low complexity, and yet high success rate in many practical examples. However, due to the finite size of the datasets, it is known that kNN is not guaranteed to converge to the optimal solution [18]. Breiman's Random Forest method [19] uses an ensemble of decision tree classifiers with random data splitting at each node, thus being robust to noise. There have been reports of its potential in imbalanced datasets even with simple sampling with replacement, and especially for artificially balanced datasets [20,21]. Support Vector Machines have been previously used for the task of labour classification [22,23]. In this study, the use of SVM with polynomial kernel was evaluated.

D. ADASYN

The problem of learning from data with skewed class distribution is relatively frequent in medical applications of machine learning [24]. The solutions to tackle the imbalanced classes are summarized in [24], and briefly divided into data and algorithm level approaches. Data level approaches are more universally applicable and do not demand in-depth knowledge of the learning algorithm [24].

An available solution at the data level is to generate artificial examples of the minority class. This is the fundamental idea for the Synthetic Minority Over-sampling Technique (SMOTE) [25], which is further developed in the Adaptive Synthetic (ADASYN) sampling approach. The latter combines the oversampling from SMOTE with boosting, creating more synthetic examples corresponding to the more challenging samples of the minority class [15].

III. RESULTS

The average duration of the EHG recordings was 29.6 min. The preprocessed signals, without the first 700 samples containing artefacts, were halved into two equal parts of an approximate duration of 15 min. In this way, every subject contributes two EHG recordings, giving in total 38 EHG recordings for the EP group, and 284 for the ET group. The features corresponding to each 15 min long EHG recording were comprised of the features extracted for all three constituting channels.

The EHG channels were decomposed using EMD into 9 modes, and their instantaneous frequencies obtained upon the Hilbert transformation were inspected. It was possible to identify the three IMFs that were consistently in the predefined frequency range of interest [0.3-1] Hz in both groups of the EHG recordings. The distribution of the median frequencies of the extracted IMF_2 , IMF_3 and IMF_4 for the ET and EP are presented in Fig. 2. Although IMF_2 is at the upper frequency limit, it was included into the evaluation as to represent the frequency ranges above 0.5 Hz (Fig. 2).

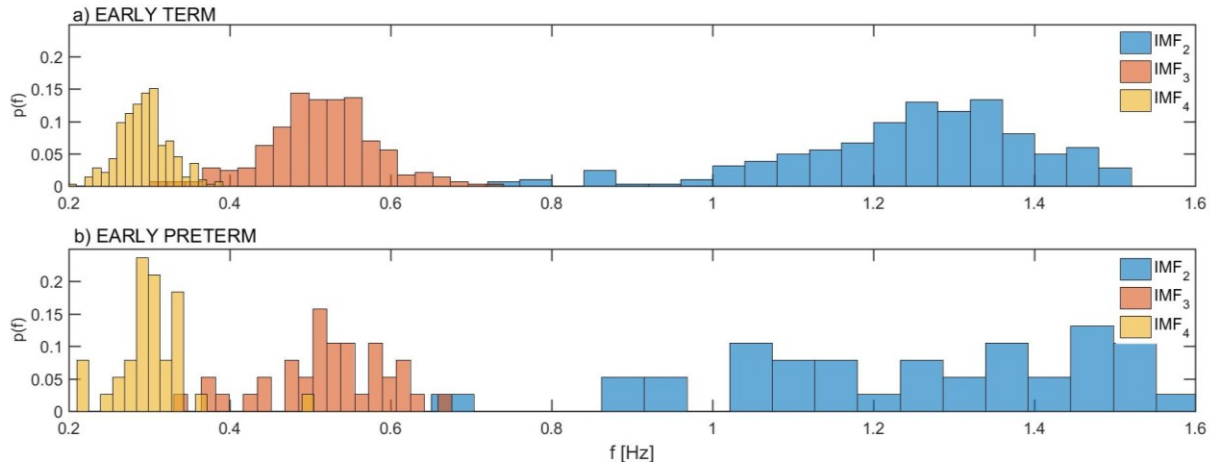


Figure 2. The distributions of the median frequencies of the selected IMFs in a) the term and b) preterm recordings. The selected IMFs cover the frequency range of interest. There is no significant difference in the corresponding distributions, as Kolmogorov-Smirnov test shows.

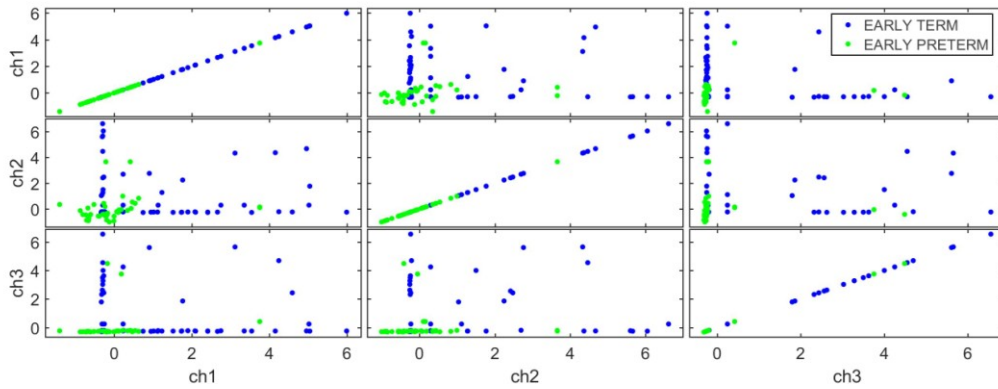


Figure 3. The scatter plots for the error of the 10th order autoregressive model in predicting channel 1 (ch1), channel2 (ch2), and channel 3 (ch3) of the term (blue) and preterm (green) recordings.

The feature selection was based on the difference in the feature distribution in the ET and EP classes, which was evaluated using the Kolmogorov-Smirnov test. The hypothesis of equal distributions was rejected in 25 features, hence providing the subset of features for the prediction task. Among those features, the following six had significantly different distributions between the ET and EP groups in all recorded channels: the error of the 10th order autoregressive model (Fig. 3), the median of the PSD over 30s segments, the power as estimated from HES

[2-3] Hz band, and the deviations of the channel variance over 30s segments.

The classification was performed with three aforementioned algorithms: k Nearest Neighbors ($k=3$), Support Vector Machine (with polynomial kernel) and Random Forest ($N_{\text{trees}}=32$). Several classifier performance metrics were observed: accuracy, precision (Pre), sensitivity for preterm class (Se), specificity (Sp), and area under curve (AUC). The preterm class was taken as the positive class; thus the sensitivity measures the success of identification of the preterm pregnancies. Since the task of predicting preterm deliveries requires minimizing the number of preterm instances classified as term class (false negatives), the classifiers were evaluated and compared primarily based on their sensitivity, but with respect to the precision. Due to the limited number of the preterm samples (around 12 % of the total number), and an overall small dataset (total of 322 recordings), the testing was performed using k -fold cross-validation, with k set to 10. Additionally, every test set had to contain samples of both classes. The 10-fold cross-validation was repeated randomly over 100 iterations, and the average of classifiers' evaluation metrics are presented in Table I. The best result is achieved by the

TABLE I.
CLASSIFICATION RESULTS

Algorithm	10-fold cross-validation			
	Pre (%)	Se (%)	Sp (%)	AUC
kNN	34.91	92.68	99.28	0.717
SVM	49.13	46.14	94.09	0.632
RF	78.55	96.62	96.80	0.964
RF+SMOTE	90.64	94.17	97.78	0.983
RF+ADASYN	95.86	98.40	99.76	0.999

Random Forest classifier. The results were additionally improved by balancing the dataset prior to RF classification using SMOTE and ADASYN. An increase in precision from 79% to 96% when ADASYN sampling is applied was noticed. In comparison with the best previously achieved result, the sensitivity is improved from 96% [4] to 98.40%, while maintaining the high AUC (99%) as a measure of an overall accuracy.

IV. CONCLUSION

This study evaluates the possibility of predicting the preterm delivery from the EHG recordings made between 22nd and 25th week of gestation, from the publicly available Term-Preterm EHG Database. A novel set of features for the detection of preterm EHG recordings was introduced and evaluated. The features capture both linear traits, using the predictive modelling, and the nonlinear, nonstationary characteristics of the uterine EMG signals using empirical mode decomposition, and monitoring the changes in the features extracted from the short stationary segments. The results indicate a significant discriminatory capacity of the error of AR modelling with 10th order polynomial (Fig. 3), and of the features measuring the standard deviation over recording time of the PSD and the variance, as estimated over 30s segments. The values of these features indicate the more predictable and less variable behaviour of the preterm EHG recordings (as noticeable in Fig. 1). The estimated features after EMD decomposition generally lack discriminatory capacity, except for the features estimated from the Hilbert energy spectra. Regarding classifiers, Random Forest classifier exhibited the best performance, further improved with ADASYN to solve the imbalanced classes problem.

While the previously published findings on the TPEHG dataset exhibited comparatively high performance, the use of the introduced features decreases the classification error and increases the sensitivity. Another contribution worth noting is reduced, halved, recording time needed to achieve this excellent classification performance. Altogether, the results indicate that non-invasive and cost-effective EHG monitoring during pregnancy helps to identify the potential pregnancies at risk of a preterm delivery, ensuring timely, preventive interventions. Bearing in mind the critical importance of this task and its immense consequences, it is essential to provide a robust and trustworthy classification.

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